

Project Design Phase

Proposed Solution Template

Date	30 June 2026
Team ID	[Team ID]
Project Name	Customer Care Registry
Maximum Marks	2 Marks

Proposed Solution — Customer Care Registry

The following information describes the proposed solution for the Customer Care Registry project.

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	- Customers must repeat their issue across disconnected channels (phone, email, chat) with no visibility into ticket status, causing frustration and delayed resolution. - Support agents and managers lack a unified view of customer history, complaint categorization, and performance data, leading to slow, inconsistent resolutions.
2.	Idea / Solution description	- Customer Care Registry is a single, role-based web platform where customers raise and track complaints, agents manage assigned tickets end-to-end, and managers/admins oversee operations through one connected system. - It covers the full complaint lifecycle: registration, AI-assisted categorization, assignment, interaction logging, resolution, feedback collection, and analytics — all backed by a knowledge base for self-service.
3.	Novelty / Uniqueness	- AI-assisted complaint categorization (via the Google Gemini API) automatically suggests category, priority, and department from the complaint text, with a deterministic keyword-based fallback so intake keeps working even if the AI service is offline. - A single system unifies ticketing, multi-channel interaction logging (call, email, chat, visit, WhatsApp, video), feedback/sentiment, knowledge base, and analytics — rather than stitching together separate tools.
4.	Social Impact / Customer Satisfaction	- Customers get transparent, real-time visibility into their complaint status and SLA due dates, reducing anxiety and repeated follow-ups. - Faster, better-categorized resolutions and a searchable knowledge base improve customer trust and overall satisfaction scores.
5.	Business Model (Revenue Model)	- SaaS subscription tiers for businesses (Standard / Enterprise / VIP support tiers) based on number of agents, ticket volume, and access to AI-assisted features. - Optional add-ons: advanced analytics/reporting, additional AI-assisted automation, and premium support/onboarding for enterprise clients.
6.	Scalability of the Solution	- Layered architecture (React UI → Express REST API → pluggable data layer) allows the current JSON file store to be replaced with a scalable database (e.g. MongoDB/PostgreSQL) without changing the API contract. - Stateless API layer can run as multiple instances behind a load balancer, and new modules (e.g. additional channels, more AI capabilities) can be added without disrupting existing functionality.